feedback loop as Build-Measure-Learn because the activities happen in that order, our planning really works in the reverse order: we figure out what we need to learn and then work backwards to see what product will work as an experiment to get that learning. Thus, it is not the customer, but rather our hypothesis about the customer, that pulls work from product development and other functions. Any other work is waste.

Hypothesis Pull in Clean Tech

To see this in action, let's take a look at Berkeley-based startup Alphabet Energy. Any machine or process that generates power, whether it is a motor in a factory or a coal-burning power plant, generates heat as a by-product. Alphabet Energy has developed a product that can generate electricity from this waste heat, using a new kind of material called a thermoelectric. Alphabet Energy's thermoelectric material was developed over ten years by scientists at the Lawrence Berkeley National Laboratories.

As with many clean technology products, there are huge challenges in bringing a product like this to market. While working through its leap-of-faith assumptions, Alphabet figured out early that developing a solution for waste thermoelectricity required building a heat exchanger and a generic device to transfer heat from one medium to another as well as doing project-specific engineering. For instance, if Alphabet wanted to build a solution for a utility such as Pacific Gas and Electric, the heat exchanger would have to be configured, shaped, and installed to capture the heat from a power plant's exhaust system.

What makes Alphabet Energy unique is that the company made a savvy decision early on in the research process. Instead of using relatively rare elements as materials, they decided to base their research on silicon wafers, the same physical substance that computer central processing units (CPUs) are made from. As CEO Matthew Scullin explains, "Our thermoelectric is the only one that can use low-cost semiconductor infrastructure for manufacturing."